



US012382886B2

(12) **United States Patent**
Marder-Eppstein et al.

(10) **Patent No.:** **US 12,382,886 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CONTROL SPACE OPERATING SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 857 days.

(21) Appl. No.: **17/182,218**

(22) Filed: **Feb. 22, 2021**

(65) **Prior Publication Data**

US 2021/0259163 A1 Aug. 26, 2021

Related U.S. Application Data

(60) Provisional application No. 62/979,364, filed on Feb. 20, 2020.

(51) **Int. Cl.**

A01G 31/02 (2006.01)
A01G 7/02 (2006.01)
A01G 7/04 (2006.01)
A01G 9/029 (2018.01)
A01G 9/24 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A01G 31/02** (2013.01); **A01G 7/02** (2013.01); **A01G 7/045** (2013.01); **A01G 9/0299** (2018.02); **A01G 9/247** (2013.01); **A01G 9/26** (2013.01); **A01G 25/09** (2013.01); **A01G 25/16** (2013.01); **A01G 27/00** (2013.01); **A01G 27/001** (2013.01); **A01G 27/003** (2013.01); **A01G 27/008** (2013.01);

A01M 7/0089 (2013.01); **B25J 11/00** (2013.01); **B60P 3/30** (2013.01); **G05B 19/042** (2013.01); **G06F 16/25** (2019.01); **G06Q 10/06315** (2013.01); **G06Q 50/02** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC **A01G 31/02**; **A01G 7/02**; **A01G 7/045**; **A01G 9/249**; **F24F 11/00**; **G05B 19/042**; **G06Q 10/06315**; **G06Q 50/02**
See application file for complete search history.

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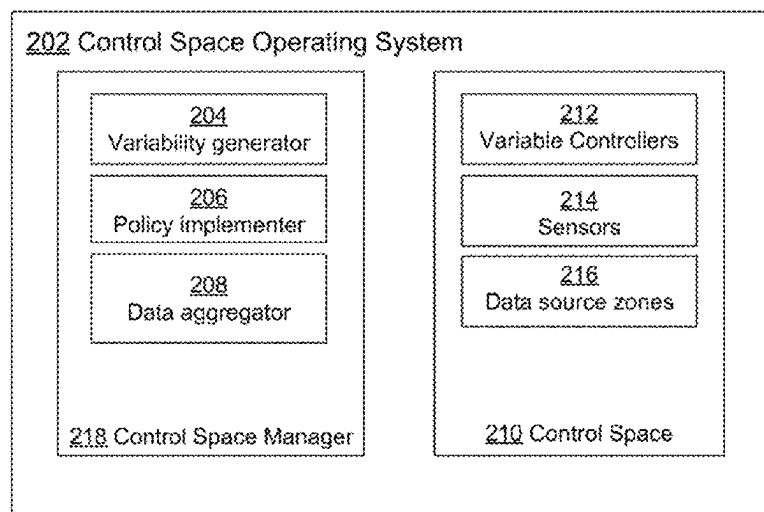
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Primary Examiner — Manuel L Barbee

(57) **ABSTRACT**

A control space operating system. The system includes a control space with one or more data source zones and a control space manager. The control space manager can collect data and control different variables across different data source zones in order to determine optimal policies and conditions for data source growth and generation.

10 Claims, 12 Drawing Sheets



(51) **Int. Cl.**

A01G 9/26 (2006.01)
A01G 25/09 (2006.01)
A01G 25/16 (2006.01)
A01G 27/00 (2006.01)
A01M 7/00 (2006.01)
B25J 11/00 (2006.01)
B60P 3/30 (2006.01)
G05B 19/042 (2006.01)
G06F 16/25 (2019.01)
G06Q 10/0631 (2023.01)
G06Q 50/02 (2024.01)

(52) **U.S. Cl.**

CPC **A01G 9/24** (2013.01); **A01M 7/0025**
 (2013.01); **G05B 2219/23133** (2013.01)

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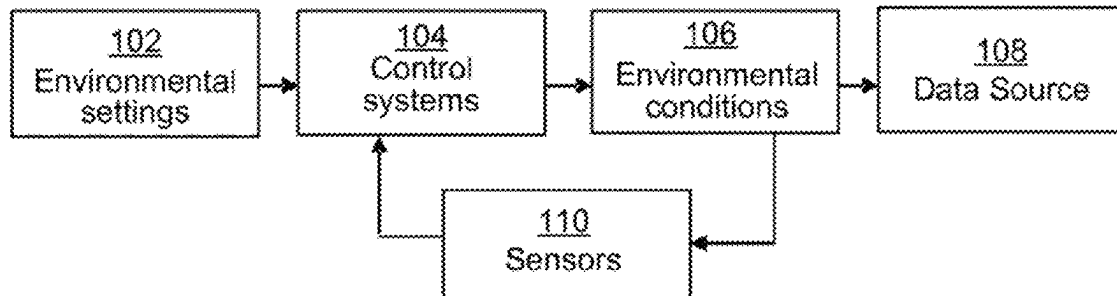
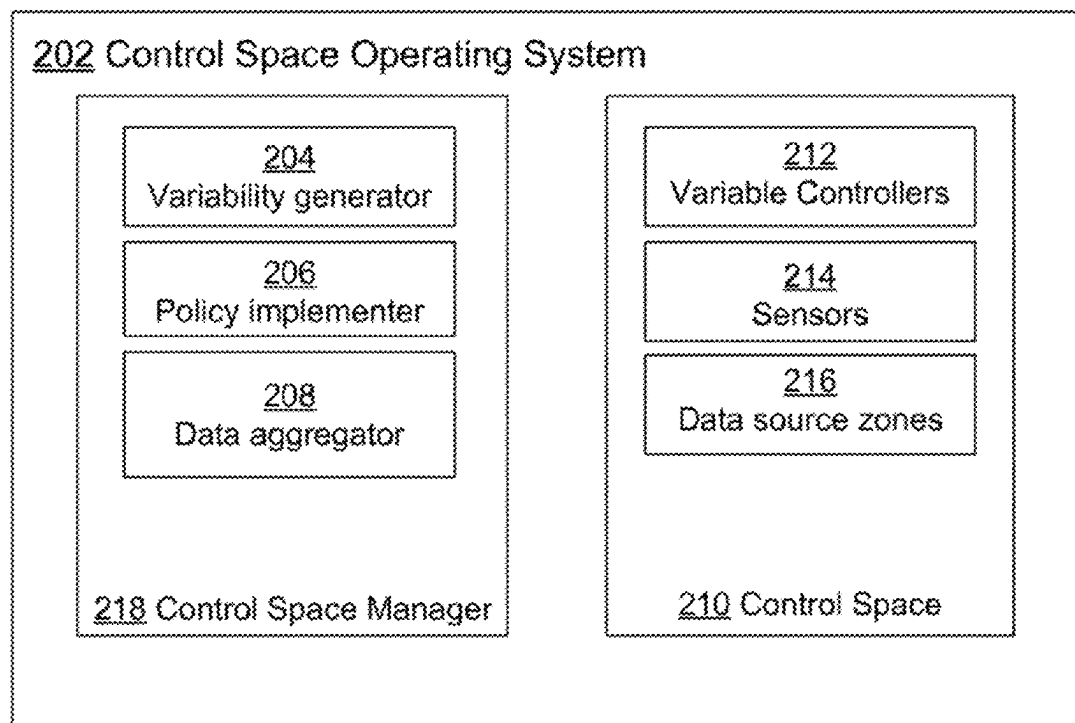
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**FIG. 1 (PRIOR ART)****FIG. 2**

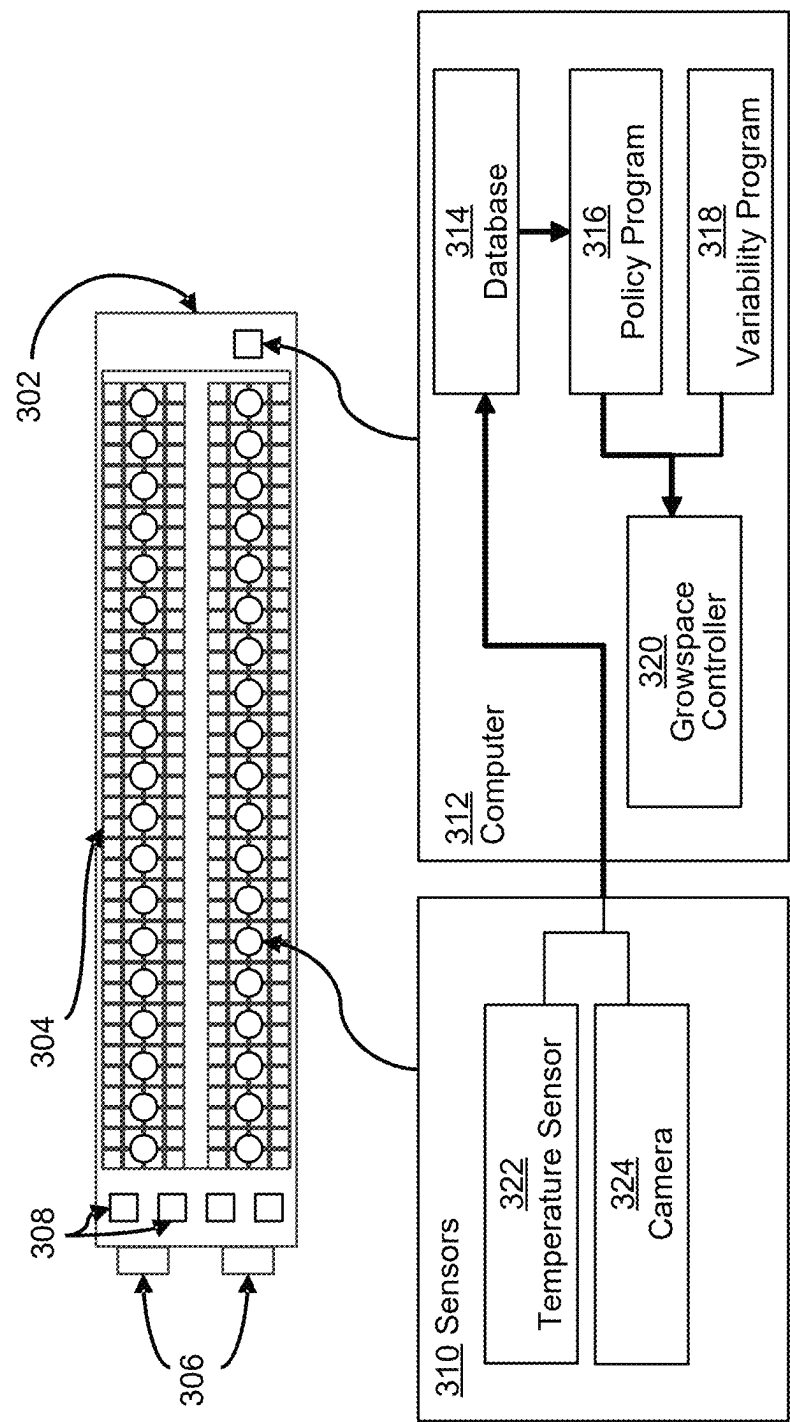


FIG. 3

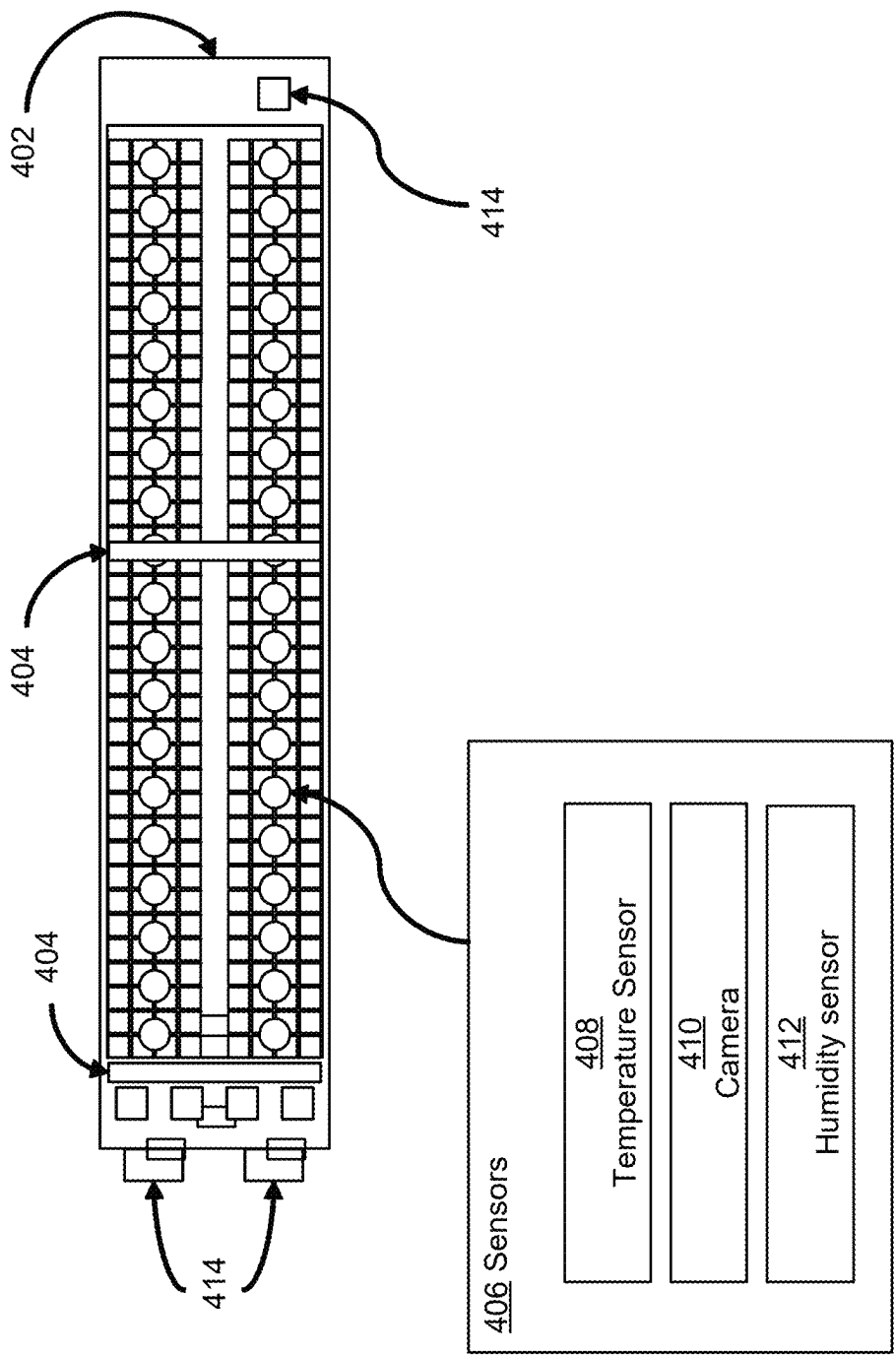
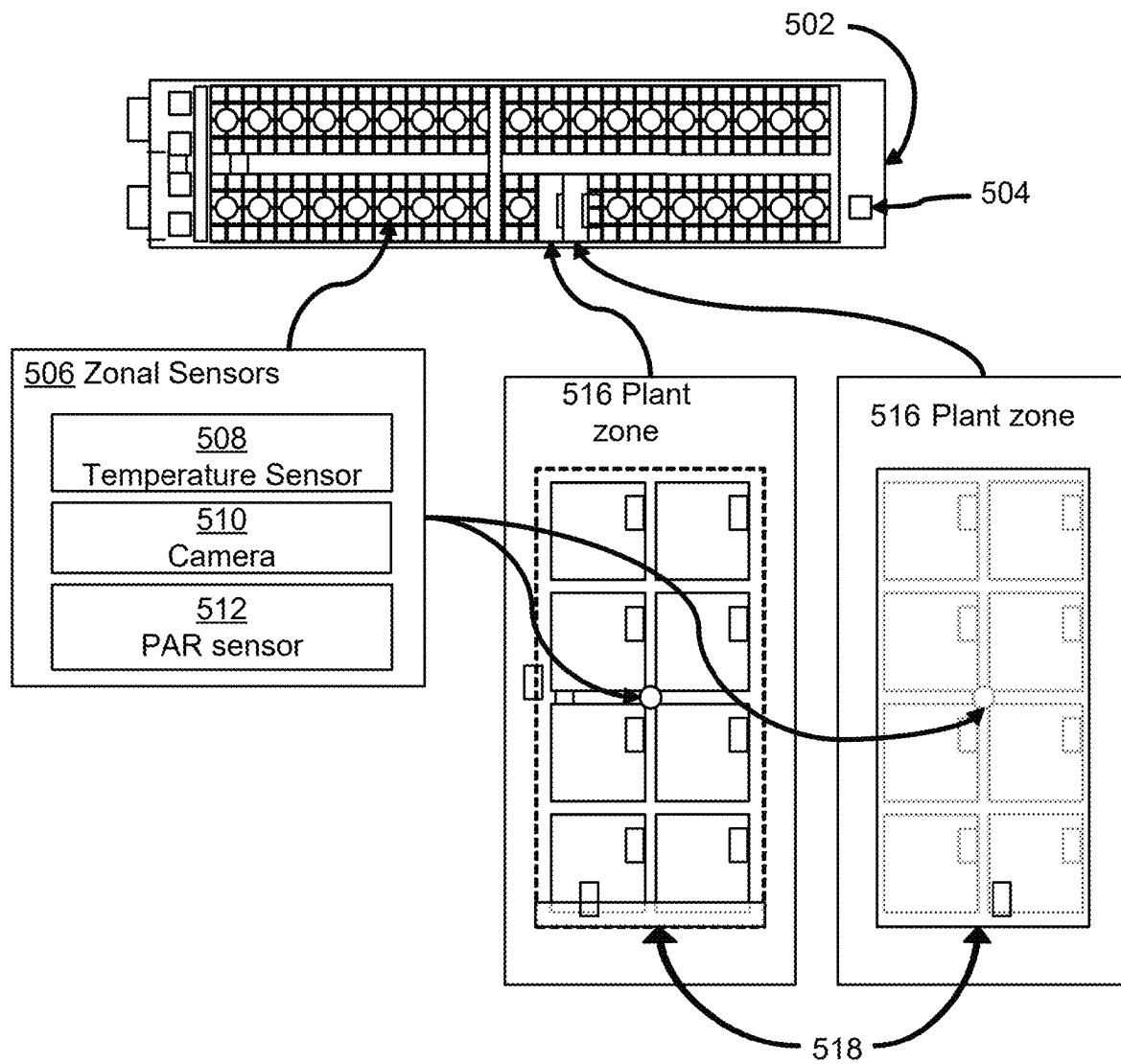
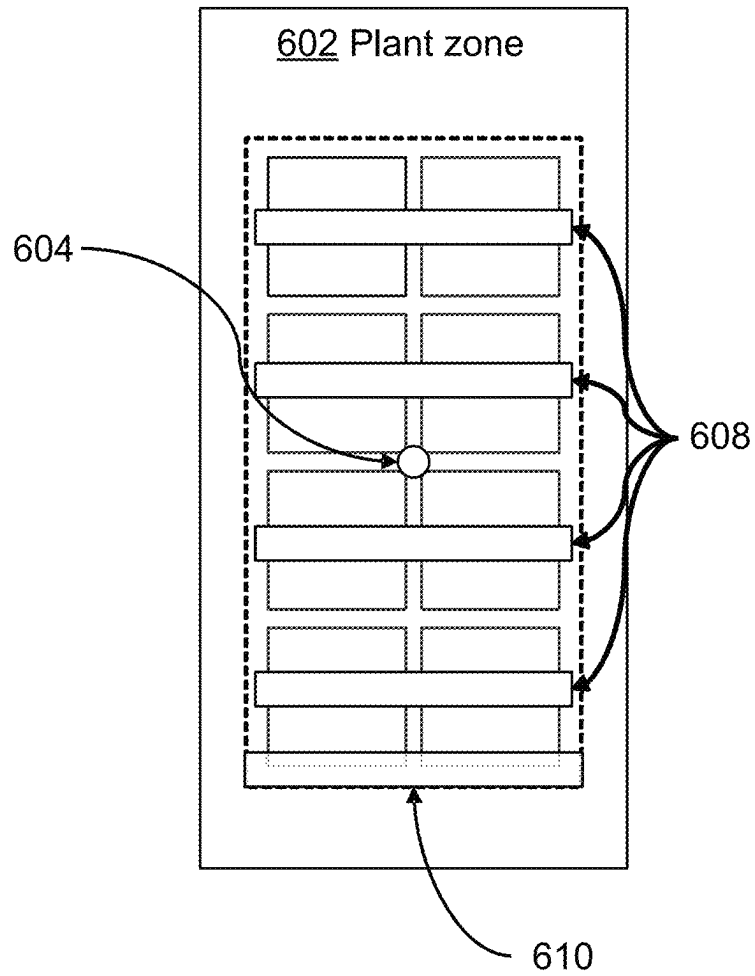
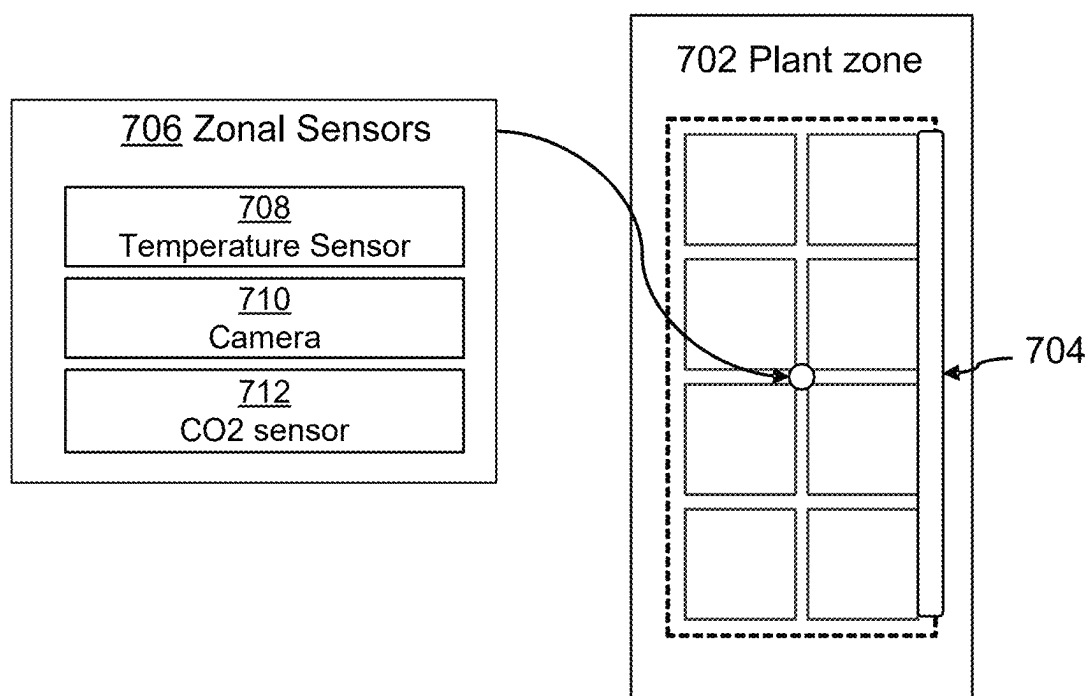
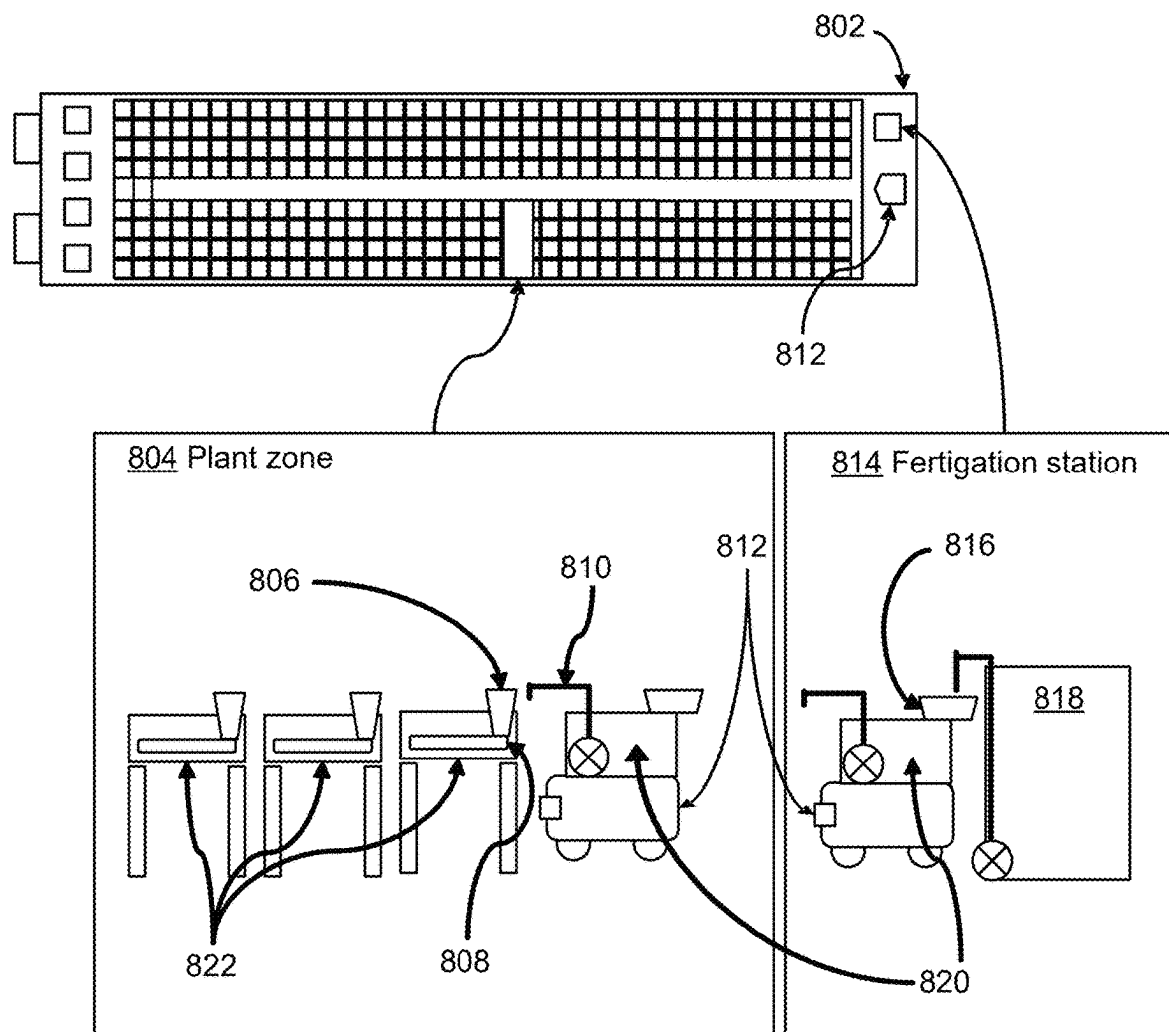


FIG. 4

**FIG. 5**

**FIG. 6**

**FIG. 7**

**FIG. 8**

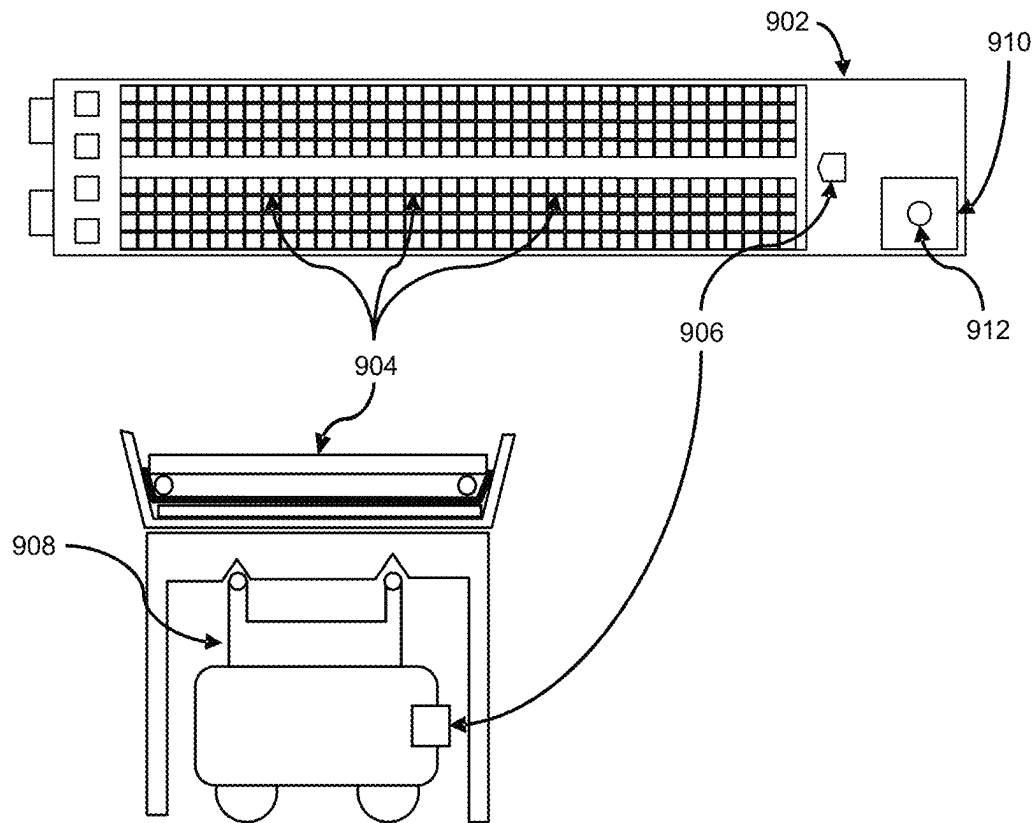
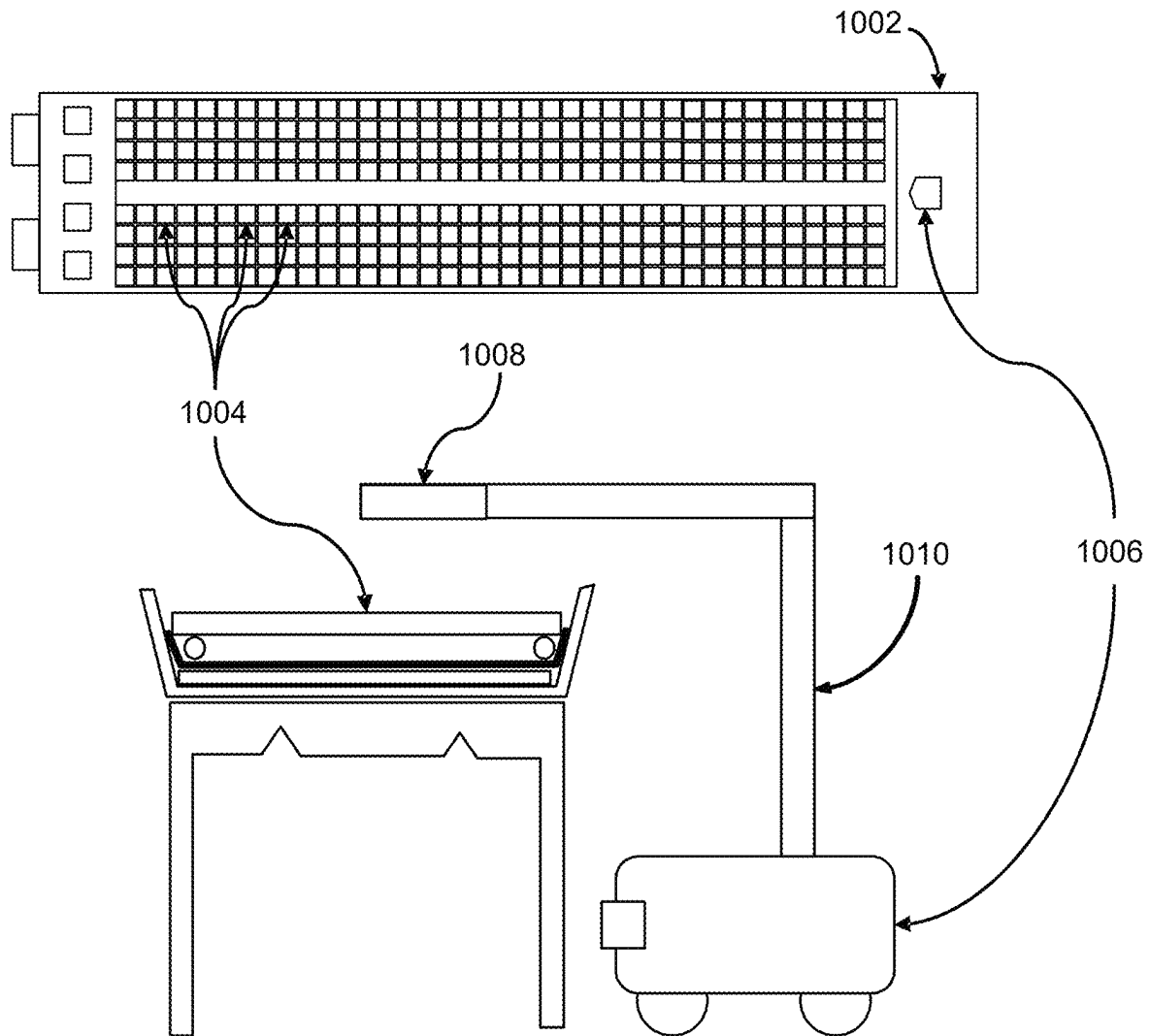


FIG. 9

**FIG. 10**

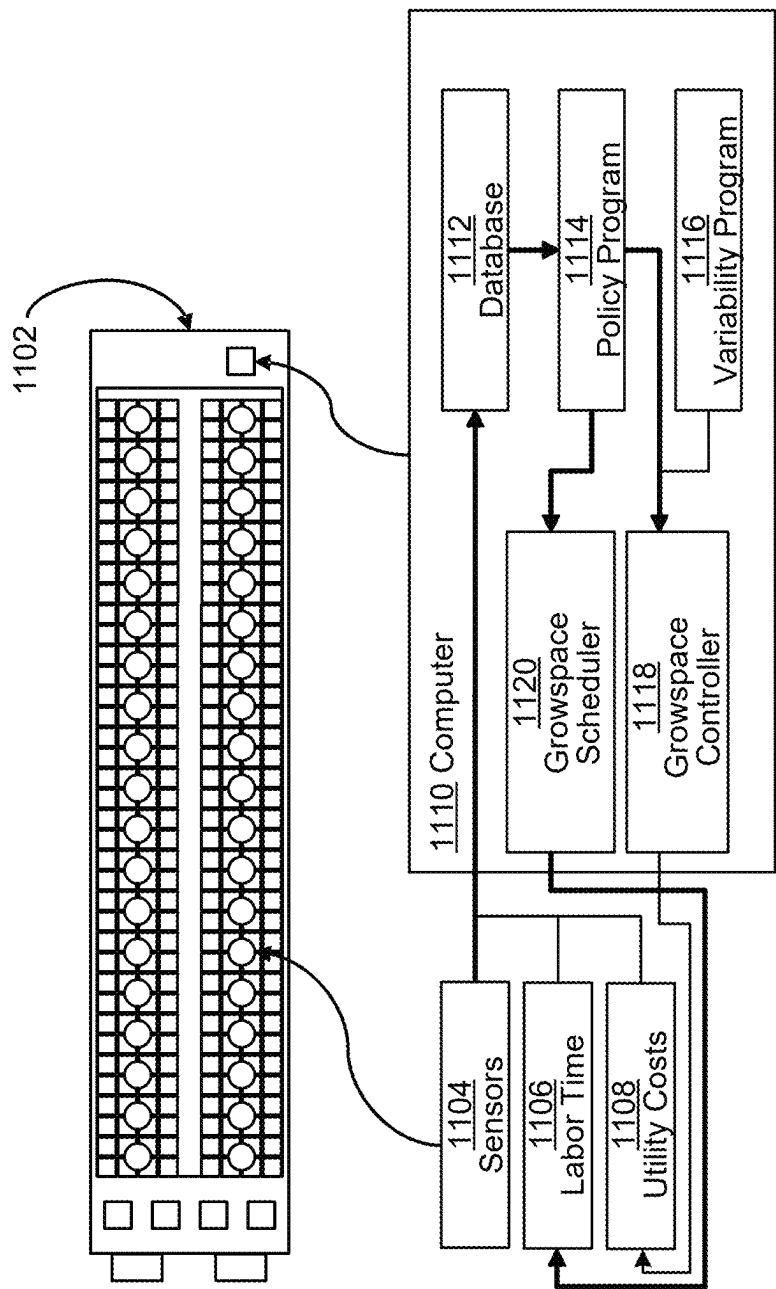
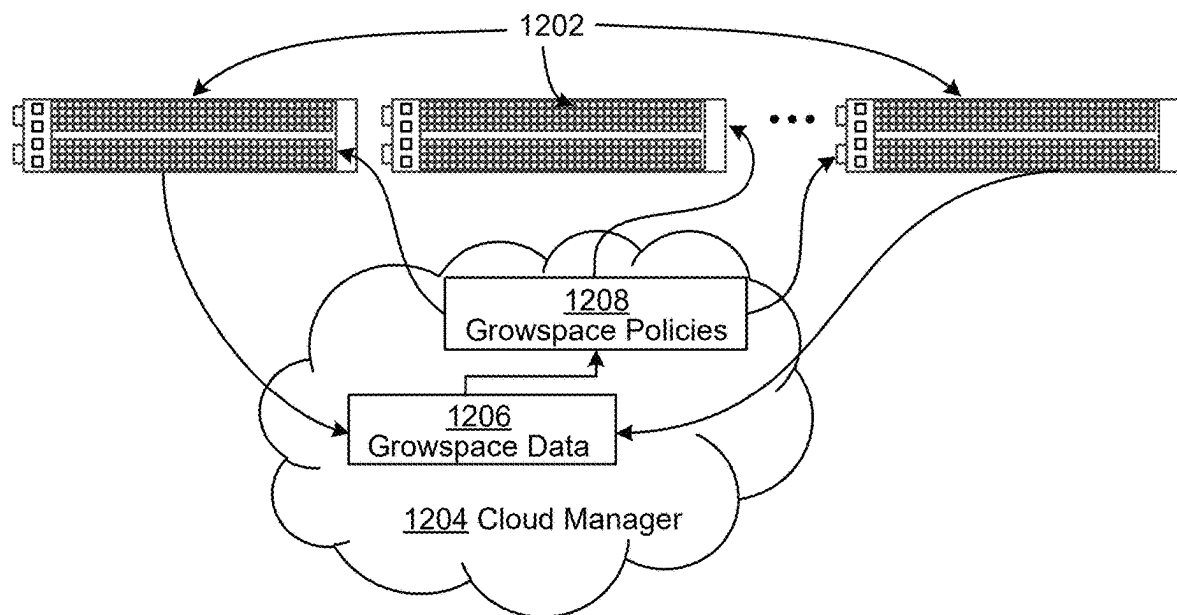


FIG. 11

**FIG. 12**

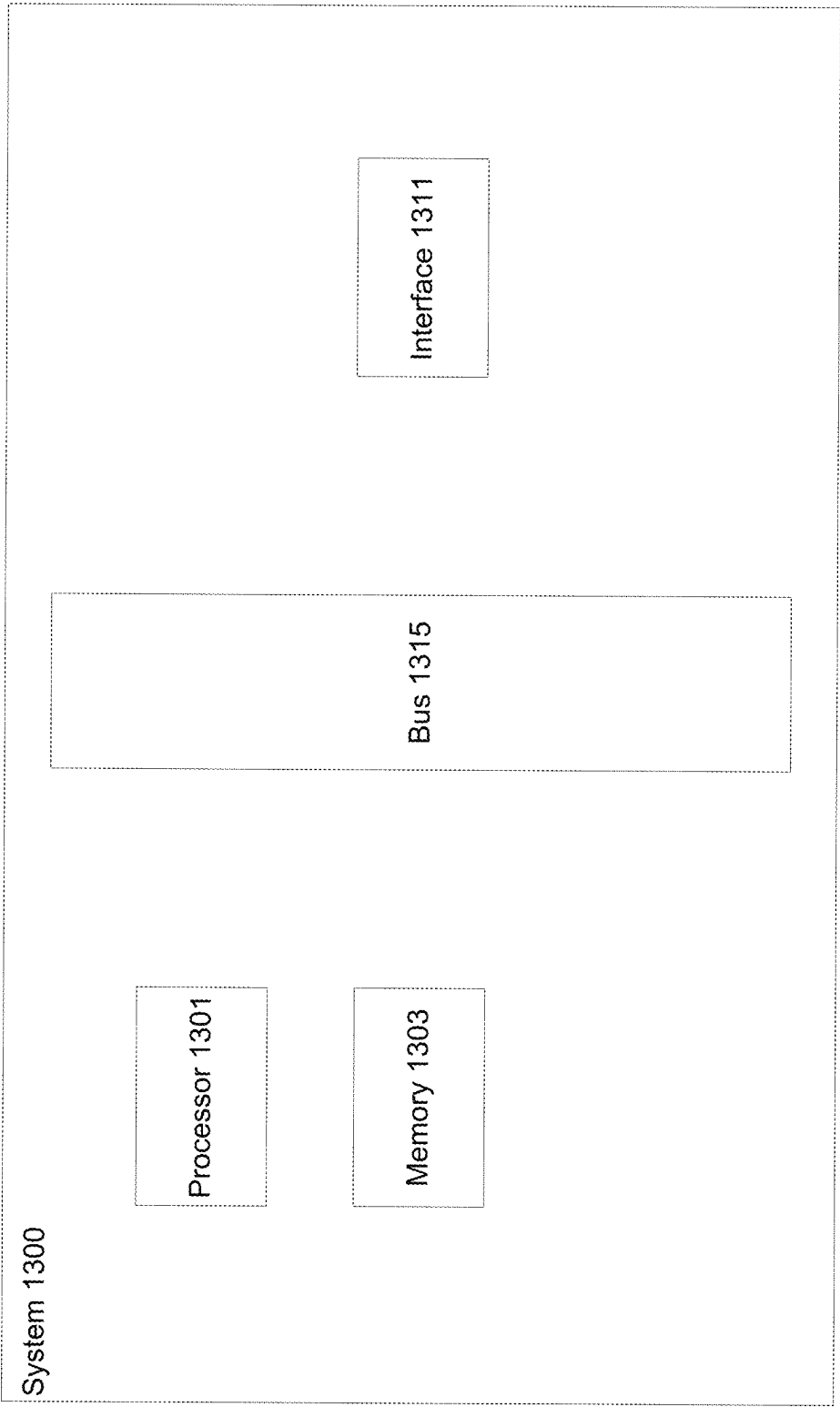


FIG. 13

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CONTROL SPACE OPERATING SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to Provisional U.S. Patent Application No. 62/979,364, titled "Growspace Operating System," filed on Feb. 20, 2020, by Eitan Marder-Eppstein et al., which is incorporated herein by reference in its entirety and for all purposes.

TECHNICAL FIELD

The present disclosure relates generally to agriculture, and more specifically to growspace farming systems.

DESCRIPTION OF RELATED ART

Agriculture has been a staple for mankind, dating back to as early as 10,000 B.C. Through the centuries, farming has slowly but steadily evolved to become more efficient. Traditionally, farming occurred outdoors in soil. However, such traditional farming required vast amounts of space and results were often heavily dependent upon weather. With the introduction of greenhouses, crops became somewhat shielded from the outside elements, but crops grown in the ground still required a vast amount of space. In addition, ground farming required farmers to traverse the vast amount of space in order to provide care to all the crops. Further, when growing in soil, a farmer needs to be very experienced to know exactly how much water to feed the plant. Too much and the plant will be unable to access oxygen; too little and the plant will lose the ability to transport nutrients, which are typically moved into the roots while in solution.

One disadvantage of traditional farming is the lack of control over the environment and growing conditions. With the advent of growspaces, external environmental factors, such as weather, can be removed. However, current growspaces are still inefficient because of the lack of modular or zonal control within a growspace. Improvements to growth are discovered through trial and error experimentation. In addition, lessons are usually learned in a research and development (R&D) facility independent from production.

Further, operating a growspace today comes with a number of challenges that place significant burdens on farmers and leads to increased costs and/or inefficient food production. For example, current growspace systems have high manual labor costs for maintenance of crops and data gathering. If farmers want to reduce labor costs, they can purchase traditional manufacturing equipment, which is very expensive. Last, current growspace systems do not have the ability to easily evolve because obtaining granular data can be infeasible and taxing on farmers.

SUMMARY

The following presents a simplified summary of the disclosure in order to provide a basic understanding of certain embodiments of the present disclosure. This summary is not an extensive overview of the disclosure and it does not identify key/critical elements of the present disclosure or delineate the scope of the present disclosure. Its sole purpose is to present some concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.

Aspects of the present disclosure relates to a control space operating system and method for growing plants using the

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control space operating system. The system comprises a control space and a control space manager. The control space includes one or more variable controllers configured for adjusting one or more variables in the control space. The control space also includes one or more sensors for gathering data. Last, the control space further includes one or more data source zones. Each data source zone is configured to house a data source. The control space manager includes a variability generator configured for determining degrees of adjustment to the one or more variables across different data source zones or for each data source zone. The control space manager also includes a policy implementer configured for determining an optimal policy for a specified criteria. Last, the control space manager further includes a data aggregator configured to collect or store data gathered from the one or more sensors.

In some embodiments, the one or more variables includes nutrient mixtures. In some embodiments, each data source zone allows full control over lighting conditions in the data source zone, independent of other data source zones. In some embodiments, each data source zone includes zonal light emitting diodes (LEDs) or zonal shades for adjusting light in each data source zone. In some embodiments, the one or more variables includes humidity. In some embodiments, the data aggregator utilizes a mobile robot to sense data. In some embodiments, the control space includes a designated centralized sensing area to which data sources are transported for sensing data. In some embodiments, the policy implementer utilizes one or more of the following data signals in determining an optimal policy: labor time, utility cost, and sensor data. In some embodiments, data gathered from the control space is transmitted to a cloud manager that aggregates data from multiple control spaces and facilitates generation of aggregated control space policies for use by the control space manager. In some embodiments, each data source zone is configured for zonal carbon dioxide (CO₂) emission control.

These and other embodiments are described further below with reference to the figures.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure may best be understood by reference to the following description taken in conjunction with the accompanying drawings, which illustrate particular embodiments.

FIG. 1 illustrates a simple diagram showing one example of a typical control space pipeline, in accordance with embodiments of the present disclosure.

FIG. 2 illustrates a block diagram showing one example of a control space operating system, in accordance with embodiments of the present disclosure.

FIG. 3 illustrates an example control space implemented as a growspace, in accordance with embodiments of the present disclosure.

FIG. 4 illustrates an example of a control space with variable humidity control, in accordance with embodiments of the present disclosure.

FIG. 5 illustrates an example of a control space with light blocking, in accordance with embodiments of the present disclosure.

FIG. 6 illustrates an example of a control space with light addition, in accordance with embodiments of the present disclosure.

FIG. 7 illustrates an example of a control space with carbon dioxide addition, in accordance with embodiments of the present disclosure.

FIG. 8 illustrates an example of a control space with nutrient control, in accordance with embodiments of the present disclosure.

FIG. 9 illustrates an example of a control space with centralized sensing, in accordance with embodiments of the present disclosure.

FIG. 10 illustrates an example of a control space with robot based sensing, in accordance with embodiments of the present disclosure.

FIG. 11 illustrates an example of a control space with additional factor tracking, in accordance with embodiments of the present disclosure.

FIG. 12 illustrates an example of a multiple control space operating system, in accordance with embodiments of the present disclosure.

FIG. 13 illustrates an example of a computer system, configured in accordance with one or more embodiments.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Reference will now be made in detail to some specific examples of the present disclosure including the best modes contemplated by the inventors for carrying out the present disclosure. Examples of these specific embodiments are illustrated in the accompanying drawings. While the present disclosure is described in conjunction with these specific embodiments, it will be understood that it is not intended to limit the present disclosure to the described embodiments. On the contrary, it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the present disclosure as defined by the appended claims.

For example, portions of the techniques of the present disclosure will be described in the context of particular computerized systems. However, it should be noted that the techniques of the present disclosure apply to a wide variety of different computerized systems. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. Particular example embodiments of the present disclosure may be implemented without some or all of these specific details. In other instances, well known process operations have not been described in detail in order not to unnecessarily obscure the present disclosure.

Various techniques and mechanisms of the present disclosure will sometimes be described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless noted otherwise. For example, a system uses a processor in a variety of contexts. However, it will be appreciated that a system can use multiple processors while remaining within the scope of the present disclosure unless otherwise noted. Furthermore, the techniques and mechanisms of the present disclosure will sometimes describe a connection between two entities. It should be noted that a connection between two entities does not necessarily mean a direct, unimpeded connection, as a variety of other entities may reside between the two entities. For example, a processor may be connected to memory, but it will be appreciated that a variety of bridges and controllers may reside between the processor and memory. Consequently, a connection does not necessarily mean a direct, unimpeded connection unless otherwise noted.

EXAMPLE EMBODIMENTS

As mentioned above, current growspace systems have many drawbacks. For example, labor costs are high (typi-

cally 60-80% of operating expenses) and reliability can be a problem at scale. It can be hard to find/retain good employees, maintain quality, and remain price competitive in an industry that often pays minimum wage or lower (e.g. migrant labor). This is especially true for growspaces that operate in urban areas with higher cost of living and minimum wage.

Another drawback can be capital expenditure. If growspaces want to reduce labor costs, they can look into automation. However, with current technology, automation to reduce labor costs is inflexible and capital intensive. Those growspaces that are automated use traditional process manufacturing techniques, e.g., conveyor belts, cart+rail, or raft systems that are expensive to install, crop specific (e.g. only work with lettuce or tomatoes, not both), and extremely difficult to reconfigure/move once put in place.

Yet another drawback is the lack of data. Getting good, granular data on crop production can be hard. Growspace farmers today struggle to answer questions like “How much labor went into this unit of produce (e.g. head of lettuce, single tomato, etc.)?”, “What operations were applied to it and when? (e.g. pest control, pruning, transplanting)”, “What is the unit cost of production for the produce we grow?” Traditional methods of tracking labor/materials often rely on immediate data entry that is challenging for farmers that are out in the field, wearing gloves, around lots of water, and unable to regularly interact with electronic devices like phones or computers while working.

Many growspaces are built without data collection in mind requiring retrofits after the fact just to be able to start collection. These retrofits are challenging and expensive as it can be hard to get sensors into a control space that provide sufficient data volume for today’s machine learning systems.

The lack of data is often compounded by the slow rate of learning. Experimentation cycles are slow. When farmers want to experiment to improve production in growspaces today they are limited by their fixed infrastructure. Process improvements, tweaks to growing methods, and modifications to growing hardware are often impossible or prohibitively expensive because they imply retooling of the entire growspace. Often, farmers will wait until they build a new growspace to make changes based on learnings from their last operation which leads to improvement cycles that take years. Often times, experimentation and data generation is separate from production. Most of the learning happens in an R&D facility and lessons learned are moved to production through a gradual process of trials. This separation leads to R&D spaces that are much smaller than production spaces and limits the numbers of experiments that can be run. Learning rates with this model are slow. In addition, data gathering in current systems require manual labeling of data. Generating these labels, even in the presence of sufficient data volume is challenging and expensive. Further, current systems struggle to track a data source through its entire lifetime and through automation pipelines. This leads to very coarse metrics (e.g. statistics on the entire production facility, but not on any one data source) that are unsuitable for generating detailed insights.

Last, one other major drawback with current growspace systems is the inability to support diversification. Growspaces that have automation built into them are only capable of growing a small set of crops (often just one) that are aligned with the tooling they have. If a growspace growing lettuce loses a major customer, but finds a replacement that wants tomatoes instead, there is no easy way to switch. The cost of retooling and effort of reconfiguring a growspace prevents growers from making that kind of change. In

addition, farmers cannot grow multiple crops or change what they grow based on the time of year or market patterns without changing automation systems (e.g. farmers cannot ramp up tomato production in the winter, but then swap it out for lettuce in the summer as field tomatoes flood the market).

The systems and techniques disclosed herein address the above mentioned issues by providing a control space operating system that utilizes robotic transport, centralized processing, and scheduling/monitoring/tracking software. According to various embodiments, a control space can be a type of grow space, but with much more control over variables.

The systems and techniques disclosed herein provide many advantages over current growspace systems. For example, in some embodiments, the disclosed automation systems are modular, requiring less up-front capital investment and allowing for gradual expansion of a grow operation. In some embodiments, the automation systems disclosed are decoupled from the crops being grown, which means that the techniques and systems work across many different crop types (e.g. lettuce, tomatoes, strawberries, etc.). In some embodiments, the automation systems disclosed are flexible and can be reconfigured on the fly, e.g., using mobile robots instead of conveyors means we can make changes to our farm in software rather than reconfiguring conveyors. In some embodiments, the automation systems disclosed allow for random access to plants. By contrast, conveyor and raft systems only allow farmers to access plants that are at the beginning or end of the conveyor or raft system. In such systems, if anything happens to plants in the middle (e.g., a disease) it's very difficult for growers to take action or even identify that the problem exists using traditional automation processes. In some embodiments, the automation systems disclosed allow for plant level tracking and data collection throughout the growth cycle with scheduling, monitoring, and management software vertically integrated into transport.

Yet another advantage is that, according to some embodiments, the control space is built specifically for data collection, as well as organizing the space, sensors, and controls together to enable large scale experimentation in production environments. Since experiments are no longer restricted to R&D settings only, that data volume scales with the size of production facilities and is not limited to the space dedicated to R&D.

Yet another advantage is that, in some embodiments, the control space is built specifically to ensure sufficient coverage of the variable space to provide neural networks with the variability/richness they need to learn how changes to environmental or other parameters impact a data source. Each data source zone is ensured of running a slightly different policy from any other at all times.

Yet another advantage is that, in some embodiments, the control space is built with automated labeling and tracking in mind. Sensors for and the structure of each data zone are designed to make the task of tracking output metrics (e.g. growth, volume, yield) a natural byproduct of daily operation which greatly reduces or eliminates the need for manually labeled data.

According to various embodiments, the control space operating systems comprises a number of distinct components/modules/subsystems that operate together. However, it should be noted that techniques of the present disclosure do not require all components/modules/subsystems described. For example, in some embodiments, a control space according to the present disclosure can include a single subsystem

or any combination of the different subsystems described herein. The different components/modules/subsystems are described in detail below.

Increasingly, data and automation are becoming important components for controlled environment agriculture (CEA) grow spaces, biotech facilities, warehouses, data centers, test spaces for experiments, and other control spaces. However, current control space architectures and their associated control systems make it difficult to introduce variability in environmental conditions that lead to a sufficiently rich understanding of how such conditions impact production conditions. This limitation leads to data pipelines that lack information richness and that are challenging to use with modern machine learning tools which require large amounts of labeled, rich, data to function. Furthermore, control space automation and control systems are frequently designed and employed independently from control space sensing which hampers the efficiency of collection.

FIG. 1 illustrates a simple diagram showing one example of a typical control space pipeline. In FIG. 1, desired environmental settings **102** are passed to control systems **104** which use sensors **110** to attempt to achieve a set of observed environmental conditions **106** for data sources **108**. The goal of such control pipelines is to ensure that every data source **108** in the control space experiences environmental conditions **106** that are as uniform and have as little variability as possible. While this achieves consistent production, it makes it hard to determine whether the environmental settings in use are optimal. Any experiments with environmental settings **102** become high risk as they impact production of the entire control space. In addition, cycle times are long, as only one experiment can be run at a time. To combat this, control space operators of today often build separate facilities for experimentation or look to findings from scientific/research institutions. However, the scale of these operations leads to insufficient data volume and the pace of innovation is slow. Allowing for more variability in control space operation at scale to provide modern machine learning tools with the data volume and richness they require can greatly increase the speed of innovation in the CEA, biotech, warehousing, data center, and other related spaces which employ environmental controls and sensors.

FIG. 2 presents a control space operating system **202**, where the core components of a control space are designed to work together to allow for flexible and effective data collection, aggregation, and processing and to capture variable, rich, and voluminous data. In system **202**, a data source **216** is produced in a control space **210** outfitted with variable controllers **212** that allow influence over the environment, and sensors **214** capable of measuring current environmental conditions, as well as the status of data source **216**. Control space **210** is paired with a control space manager **218**, which is the mechanism by which sufficient data volume, data richness, and policy control are achieved to support advanced machine learning techniques including the training and use of neural networks in control space operations. One example of a control space is a growspace for CEA. In other examples, the control space is a test space or experimental space used to run tests or experiments. In yet other examples, the control space is a data center, biotech production facility or warehouse.

According to various embodiments, in order to ensure data richness and volume, control space manager **218** employs a variability generator **204** that works in conjunction with variable controllers **212** that are specifically designed to have the ability to introduce variability in

environmental conditions that data source zones **216** experience across the control space **210**. In some embodiments, each data source zone **216** is configured to hold one or more data sources. In some embodiments, this data source is plants. In some embodiments, data sources are bacterial or other biological material. In some embodiments, data sources are servers. In some embodiments, data sources are any type of experimental subjects. In some embodiments, data sources are hardware that must operate under different conditions.

In some embodiments, variability generator **204** modifies variable controller **212** settings to run many parallel experiments across control space **210** to determine how data source production is impacted by environmental parameters. In some embodiments, these parameters include temperature, light, humidity, nutrients, oxygen, carbon dioxide, genetics, etc. In some embodiments, each experiment is tracked by sensors **214** in control space **210** and evaluated by data aggregator **208**, which uses machine learning to build a detailed understanding of data source production based on the factors listed above.

According to various embodiments, insights from data aggregator **208** give policy implementer **206** information that can be used to implement or generate new policies. These new policies determine variable settings for data source zones **216** that optimize for volume, production cost, variability, or other desired outcomes for production in control space **210**. In some embodiments, these settings determine starting points for control space **210** configuration, variable controllers **212**, and data source configurations that are passed to variability generator **204** to refine its exploration of the parameter space on promising areas.

According to various embodiments, the work of control space manager **218** components creates a strong feedback loop wherein large amounts of distinct data points or experiments on data source production are generated in parallel. In some embodiments, this data is used to build a detailed understanding of how data source production is impacted by variable settings. In some embodiments, that understanding is used to predict promising policy settings for variables according to a desired optimization criteria. In addition, these predictions are used and perturbed to generate more data focused on an encouraging area of the variable search space. In some embodiments, this feedback loop is the mechanism by which improvements to control space performance can be greatly accelerated compared to approaches employed today.

A specific implementation of the general system described above, is shown in FIG. 3. FIG. 3 illustrates an example control space implemented as a growspace **302**. In other words, the control space is embodied by a growspace for plant production in controlled environment agriculture (CEA). In FIG. 3, a growspace **302** is equipped with fans **306** and heaters **308** that can be used to modify the temperature in which plants **304** are grown.

According to various embodiments, when cooling is desired, fans **306** move cool air from outside growspace **302** through the structure creating a temperature gradient where air is cooler closer to the fan side of growspace **302** compared to the opposite side of growspace **302**. The slope of this gradient (e.g. the difference between the temperature close to and opposite the fans) is determined by the speed at which fans **306** move air through growspace **302**. When fans **306** move air slowly, there is more opportunity for radiant energy (e.g. from the sun) to heat air as it moves through growspace **302**, leading to a larger temperature gradient across growspace **302**. When the fans move air quickly,

there is less opportunity for air to heat up leading to a smaller temperature gradient across growspace **302**. As such, variability generator **204** can introduce more or less variability in temperature by changing the speed of fans **306**.

According to various embodiments, when heating is desired, heaters **308** move hot air created by burning natural gas, propane, or other means through growspace **302**. The temperature gradient of air across growspace **302** is, once again, impacted by the speed at which heaters **308** output air. If the heaters output air slowly, there is more time for air to lose heat as it moves from the heater side of growspace **302** to the opposite side, leading to a larger temperature gradient. If the heaters output air quickly, there is less time for air to lose heat as it moves from one side of growspace **302** to the other leading to a smaller temperature gradient.

According to various embodiments, sensors **310** placed amongst the plants **304** are spread throughout the growspace and monitor observed conditions for an area of growspace **302**, while logging their readings to a computer or group of computers **312**, which may be located on site or remotely. In some embodiments, these sensor readings are then sent to database **314** where they are stored for later processing. In some embodiments, temperature sensors **322** are used to record the temperature that plants **304** experience in their region of growspace **302**, while cameras **324** are used to collect imagery of plant growth over time.

According to various embodiments, once data on a full growth cycle, from seeding to harvest, is collected for a plant **304**, policy program **316** pulls associated data from database **314** for processing. Policy program **316** computes growth curves for plants from imagery taken by camera **324** and associates this with data from temperature sensor **322**. Policy program **316** repeats this process for growth cycles of all plants **304** that have been grown to the current point and compares results, optionally with human input, to determine temperature settings for growspace **302** that are likely to optimize plant growth.

According to various embodiments, these temperature settings are output from policy program **316** and passed to growspace controller **320** which is responsible for controlling fans **306** and heaters **308** within growspace **302** to achieve desired environmental conditions. In addition to these settings, growspace controller **320** also takes input from a variability program **318** that outputs a desired variability in temperature range for growspace **302** (e.g., it requests a 10 degree difference from one side of the grow-space to another). In some embodiments, separating policy generation and implementation and desired experimental variability into two separate components is the mechanism by which learning rates in a growspace are greatly accelerated compared to current approaches. Specifically, this decoupling explicitly pursues the variability required for neural networks to effectively explore the impact of environment on plant performance. Traditional growspaces may concern themselves with policy implementation, but not in ensuring the data they generate in production is compatible and effective with modern machine learning techniques. As such, they often lack sufficient data richness and variability for these techniques to be effective.

According to various embodiments, growspace controller **320** combines the temperature settings specified by policy program **316** with the desired variability expressed by variability program **318** to determine the speed at which to run fans **306** for cooling or heaters **308** for heating. As described above, the air speed of fans **306** or heaters **308** will determine the range of temperatures that plants **304** experience.

rience in a growspace **302** centered around the base temperature settings requested by policy program **316**.

According to various embodiments, as the number of growth cycles for plants **304** increases, the system allows policy program **316** to receive data from sensors **310** that contains enough variability (as tuned with variability program **318**) to continuously improve an understanding of plant growth as it relates to temperature. This represents a large increase in data richness as compared to industry operations today, and leads to more rapid learning, insights, and tuning of a growspace **302**.

According to various embodiments, in addition to temperature, humidity plays an important role in plant growth. The example system presented in FIG. **3** does not provide a mechanism to control humidity within a growspace and typical growspace humidity controls suffer from the same problems of traditional temperature controls in that they do not optimize for variability and data richness. Thus, it may be desirable to expand the system presented in FIG. **3** such that it is also capable of providing humidity control that can be varied over the growspace to facilitate experimentation and learning via data pipelines.

FIG. **4** presents a system configuration that adds evaporative foggers **404** to growspace **402** which add humidity to the air. In some embodiments, the mechanism used to achieve this in FIG. **4** is to spray water at high pressures into the air with evaporative foggers **404** creating a fine mist that quickly evaporates in the presence of heat. The phase transition from water into water vapor is an endothermic process that increases the humidity of the air while also cooling it. In some embodiments, to control variability of humidity across growspace **402**, the fans' speeds can be used once again to determine how quickly water vapor moves from one side of the growspace to the other. A higher fan speed will decrease the differences in humidity from one side of the growspace to the opposite. A lower fan speed will lead to an increased gradient and associated difference.

According to various embodiments, in addition to evaporative foggers **404**, the system configuration presented here also adds a humidity sensor **412** in addition to temperature sensor **408** and camera **410**. In some embodiments, humidity sensors **412** spread throughout growspace **402** take localized readings of humidity that are used to report observed conditions to computer **414**. This additional data can then be taken into account by policy program **316** and variability program **318** as they determine desired environmental settings and build a detailed understanding of how humidity and temperature impact plant growth. In some embodiments, growspace controller **320** is also updated to allow control of evaporative foggers **404** in conjunction with fans **414** so that it can achieve desired settings for humidity and temperature across growspace **402** in accordance with the request of the variability and policy programs.

According to various embodiments, light is another important parameter that impacts plant growth within a growspace. In some growspace configurations, e.g., greenhouses, light enters the growspace naturally in the form of sunlight. While this provides a natural energy source for plant growth which can be economically beneficial, it can also be something that is necessary to reduce. For example, there are situations where plants receive too much light. In some embodiments, the system can control the reduction of light within a growspace in a fashion that also allows variability and richness of data across the growspace.

FIG. **5** presents an embodiment of the system that allows for light to be blocked within growspace **502** in a way that supports variation from location to location and which can

be used to further data richness. To achieve this, growspace **502** is separated into distinct plant zones **516** which contain groups for plants that will experience similar environmental conditions. The greater the number of plant zones **516** in a growspace, the more variability that can be achieved in the footprint. Each plant zone **516** has its own zonal sensors **506** to measure observed conditions. Specifically, each zone has a temperature sensor **508**, camera **510**, and a photosynthetically active radiation (PAR) sensor **512**. PAR sensor **512** measures photosynthetic light levels in the air and is used to understand how much light plants in a plant zone **516** have received over time.

According to various embodiments, when it is desirable to remove light from a plant zone **516** in accordance with a control policy produced by the components running on computer **504** as described in previous embodiments, zonal shades **518** installed in each plant zone **516** can be automatically extended or retracted. Zonal shades **518** block a percentage of light that enters plant zone **516** by blocking it with shade cloth thereby decreasing the amount of light received by plants in the plant zone. As each zonal shade **518** is controlled separately from others in growspace **502**, they provide a mechanism by which light levels can be changed in one plant zone **516** independent from any other. This, in turn, provides a mechanism for variability program **318**, described in FIG. **3** above, to ensure sufficient data richness from light removal across growspace **502** when the sun provides light input to growspace **502**.

According to various embodiments, data from the PAR **512** sensor is fed to computer **504** in addition to the other zonal sensors **506** to which allows policy program **316** to build a model of how temperature and light impact plant growth, which can be used to further improve growspace performance.

According to various embodiments, in certain growspaces where the sun is not present or the amount of sunlight in a day is not sufficient for growth, it is desirable to be able to add light into the growspace.

FIG. **6** presents an embodiment of the system that adds zonal LEDs **608** to each plant zone **602** as a mechanism to add light to a growspace. Each zonal LED **608** can be controlled separately from zonal LEDs **608** in other plant zones **602** which allows for variability and data richness across the growspace. PAR sensor **512** described in FIG. **5** above is also sufficient to monitor and manage control of zonal LEDs **608** and the combination of zonal shades **610** with zonal LEDs **608** allows for full control over the lighting conditions within a growspace. When less light is desired, zonal shades **610** can be extended. When more light is desired, zonal LEDs **608** can be turned on.

Carbon dioxide (CO₂) is a necessary component for plant growth. There is a naturally occurring amount of CO₂ in the atmosphere that is available for plants to take up, but that may not be sufficient to sustain optimal growth. Thus, it may be desirable to develop mechanisms for actively increasing CO₂ concentrations in a growspace to achieve optimal performance.

FIG. **7** presents an embodiment of the system that adds zonal CO₂ emitters **704** to each plant zone **702**. These zonal CO₂ emitters distribute carbon dioxide that is stored in compressed form or collected as a bi-product of heating the growspace and release it into the air via nozzles. Each zonal CO₂ emitter **704** is controlled independent from any other in the growspace, which allows for CO₂ to be distributed in a targeted fashion per plant zone **702**. To ensure sufficient variability and localized control over CO₂ levels, growspace controller **320** coordinates the use of growspace fans with

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zonal CO₂ emitters. Specifically, zonal CO₂ emitters are used only when the fans are off to guarantee that CO₂ distributed to a given plant zone **702** can be absorbed by its associated plants. To measure the amount of CO₂ present in a plant zone **702**, a CO₂ sensor **712** is added to a temperature sensor **708** and camera **710**, which make up the zonal sensors **706** for that plant zone **702**. This provides yet another input for computer **504** to use as it builds a detailed understanding of environmental factors and their impact on plant growth.

Nutrition is another important component of plant growth. In current growspace systems, however, it is not possible to vary nutrient mixes given to plants across the growspace as standard hydroponic plumbing systems only allow recirculation of one nutrient mixture at a time across a growspace. To better understand and optimize the impact of nutrition on plant growth, it may be necessary to increase the number of different nutrient mixes that can be deployed to plants throughout the growspace at a given time.

FIG. **8** presents an embodiment of the system that allows for nutrients to be sent to a given plant zone via a robotic plumbing system. Specifically, a robot **812** is responsible for moving nutrient water created by a fertigation system **818** within a growspace **802**. Robot **812**, goes to a fertigation station **814** located in growspace **802** where fertigation system **818** pumps nutrient water of a given composition (either pre-mixed or created on demand) into robot inflow **816** which flows into robot reservoir **820**. With nutrient water now stored for transport, robot **812** navigates to a given plant zone **804** within growspace **802**. Each plant zone **804** contains one or more growing trays **822** in which plants sit. Each growing tray **822** has a growing tray reservoir **808** which provides nutrient water to the plants in that growing tray **822**. When robot **812** arrives at a given growing tray **822** in a given plant zone **804**, it pumps water out from robot reservoir **820** through robot outflow **810** into growing tray inflow **806** which feeds growing tray reservoir **808** for a single growing tray **822**.

According to various embodiments, the ability to move a unique mix of nutrient water from a fertigation system **818** to any growing tray **822** in a plant zone **804** allows nutrients to be tailored to a specific plant zone **804** or even a single growing tray **822** within growspace **802**. This greatly increases the level of control and amount of experimentation that can be performed relative to standard hydroponic systems which can only deliver a single nutrient mix per run of plumbing. Achieving such control with traditional plumbing systems is impractical and costly as it requires separate plumbing runs per growing tray **822** coupled with complex control valves to change the flow of water throughout growspace **802**. Using robot **812** for nutrient water transport removes the need for plumbing from growspace **802** altogether while providing a high level of control over what plants receive what nutrients. This allows variability program **318** and policy program **316** on computer **312** to experiment with unique nutrient mixes per growing tray **822** that also change over time (e.g. a different nutrient mix could be delivered on day 10 of growth as compared to day 11).

The embodiments presented above rely on distributed sensors placed throughout a growspace to record data on environmental conditions as well as plant growth. However, the camera sensors (2D, 3D, multi-spectral, etc.) used to measure plant growth are often expensive and it may be prohibitive to deploy them throughout an entire growspace on cost alone. Furthermore, deploying such sensors through a growspace requires other infrastructure like reliable network connectivity and leads to many different potential

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points of failure which must be carefully monitored. Therefore, it is desirable to reduce the number of sensors that must be deployed to track plant growth and to perform sensing in a central location.

The example system configuration presented in FIG. **9** facilitates central sensing by transporting growing trays **904** with plants to a central sensing area **910** with a robot **906**. To achieve this, robots **906** are outfitted with robot lifts **908** that can pick up growing trays **904** for transport within a growspace **902**. When data on plant growth is desired, robot **906** moves a selected growing tray **904** to sensing area **910** where sensors **912** (e.g. 2D cameras, 3D cameras, LiDAR, etc.) take measurements of the plants within a growing tray **904**. As all sensing on plant growth happens in sensing area **910**, as opposed to performing sensing out in growspace **902**, the number of expensive sensors required is drastically reduced. Furthermore, sensing area **910** can be configured to provide the optimal environment for taking sensor readings of plants (e.g. with custom lighting) to ensure uniformity of sensor readings over time.

According to various embodiments, sensing requires either distributed sensors placed throughout the growspace or robot transport of plants in growing trays to a central sensing area. For systems that require distributed sensing, cost and complexity of the sensing system is high. For systems that move plants with robots, many robots are required at large growspace scales to perform sensing tasks as each sensor reading requires moving plants through a growspace for a sensor reading and then transporting them back to their original location. In environments where sensor readings on plant growth are desired frequently, it is desirable to have a sensing configuration that avoids many distributed sensors, but is also time efficient.

FIG. **10** presents a system configuration in which a robot **1006** is outfitted with sensors **1008** attached to it via a sensor fixture **1010**. Robot **1006** navigates through a growspace **1002** to place its sensors **1008** over growing trays **1004** located throughout the environment. Sensors **1008** then take readings of each growing tray **1004** and store them for processing. This configuration avoids the need for many distributed sensors to be placed in a growspace **1002**, while putting them onto a robot instead which allows systems to save on both cost and system complexity. It also removes the need to transport growing trays **1004** directly in order to perform sensing by bringing sensor **1008** to growing trays **1004** via robot **1006** making sensing a time efficient process.

Many growspaces focus on ensuring sufficient variability and richness of environmental data on plants grown within a growspace in order to use the data to optimize production according to a desired criteria, like yield or taste. However, it may also be desirable to optimize for cost, energy, or labor of production where additional data is required to allow for optimal policy selection. Specifically, data on labor costs associated with production must be measured and combined with measured energy costs of growspace controls to determine the cost per unit weight, labor per unit weight, or energy per unit weight of plant produced.

FIG. **11** shows an example system that tracks labor time **1106** and utility costs **1108** in addition to data from sensors **1104**. These three data signals are fed into a database **1112**, which gives policy program **1114**, running on computer **1110**, vital information about the likely cost of production for a given policy. In some embodiments, to gather data during operations, a growspace scheduler **1120** determines the labor required based on the current policy program **1114** and automatically times all labor operations (automated or human) that occur in growspace **1102** via computer **1110**. In

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addition to this, the actions of growspace controller **1118** are monitored to determine utility costs **1108** of a given policy program **1114** with its associated variability program **1116**. Adding labor time **1106** and utility costs **1108** to sensors **1104** deployed throughout growspace **1102** leads to a holistic view of plant production and new options for optimization (cost, energy, and labor) that are not possible in previous embodiments.

According to various embodiments, a policy program is used to optimize a growspace according to a desired optimization criteria. However, it may be desirable to gather data from and optimize multiple growspaces together to create richer and more robust models of operation. Additionally, it may be desirable to have a growspace in one location able to learn from data from growspaces in other locations.

FIG. 12 presents a system configuration that sends grow-space data **1206** from one or more growspaces **1202** to a cloud manager **1204** responsible for aggregating data across multiple growspaces **1202**. Growspace data **1206** is then used to computer growspace policies **1208** that are passed back to each growspace **1202** for execution. This configuration allows the system to scale to any number of growspaces **1202** where each growspace **1202** also benefits from the data gathered by others in its growspace network.

The examples described above present various features that utilize a computer system or a robot that includes a computer. However, embodiments of the present disclosure can include all of, or various combinations of, each of the features described above. FIG. 13 illustrates one example of a computer system, in accordance with embodiments of the present disclosure. According to particular embodiments, a system **1300** suitable for implementing particular embodiments of the present disclosure includes a processor **1301**, a memory **1303**, an interface **1311**, and a bus **1315** (e.g., a PCI bus or other interconnection fabric). When acting under the control of appropriate software or firmware, the processor **1301** is responsible for implementing applications such as an operating system kernel, a containerized storage driver, and one or more applications. Various specially configured devices can also be used in place of a processor **1301** or in addition to processor **1301**. The interface **1311** is typically configured to send and receive data packets or data segments over a network.

Particular examples of interfaces supported include Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, and the like. In addition, various very high-speed interfaces may be provided such as fast Ethernet interfaces, Gigabit Ethernet interfaces, ATM interfaces, HSSI interfaces, POS interfaces, FDDI interfaces and the like. Generally, these interfaces may include ports appropriate for communication with the appropriate media. In some cases, they may also include an independent processor and, in some instances, volatile RAM. The independent processors may control communications-intensive tasks such as packet switching, media control and management.

According to various embodiments, the system **1300** is a computer system configured to run a control space operating system, as shown herein. In some implementations, one or more of the computer components may be virtualized. For example, a physical server may be configured in a localized or cloud environment. The physical server may implement one or more virtual server environments in which the control space operating system is executed. Although a particular

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computer system is described, it should be recognized that a variety of alternative configurations are possible. For example, the modules may be implemented on another device connected to the computer system.

In the foregoing specification, the present disclosure has been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present disclosure.

What is claimed is:

1. A control space operating system, the system comprising:

a control space comprising:

one or more variable controllers configured for adjusting one or more variables in the control space;
one or more sensors for gathering data; and
one or more data source zones, each data source zone configured to house a data source that provides data to the one or more sensors; and

a control space manager configured to:

based on a desired degree of variability in data to be obtained from the one or more data source zones, adjusting values of one or more control space variables to introduce different degrees of variability across different data source zones, further comprising introducing greater variability for at least one data source zone compared to at least one other data source zone, including generating a plurality of datasets exhibiting a degree of variability that supports one or more machine learning models; and
collect or store data gathered from the one or more sensors.

2. The system of claim 1, wherein the one or more variables includes nutrient mixtures.

3. The system of claim 1, wherein each data source zone allows full control over lighting conditions in the data source zone, independent of other data source zones.

4. The system of claim 1, wherein each data source zone includes zonal light emitting diodes (LEDs) or zonal shades for adjusting light in each data source zone.

5. The system of claim 1, wherein the one or more variables includes humidity.

6. The system of claim 1, further comprising utilizing a mobile robot to sense data.

7. The system of claim 1, wherein the control space includes a designated centralized sensing area to which data sources are transported for sensing data.

8. The system of claim 1, further comprising utilizing one or more data signals in determining an optimal policy for determining variable settings for data source zones, wherein the one or more data signals include labor time, utility cost, or sensor data.

9. The system of claim 1, wherein data gathered from the control space is transmitted to a cloud manager that aggregates data from multiple control spaces and facilitates generation of aggregated control space policies for use by the control space manager.

10. The system of claim 1, wherein each data source zone is configured for zonal carbon dioxide (CO₂) emission control.

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